

Ex.No.7 Use AFLogicalOSE to extract data from an Android device

Description

AFLogical OSE (Open Source Edition) is a tool used to extract data from Android devices. It's part of the Open Source Android Forensics project and is often used for logical extraction, which means pulling data like contacts, messages, and call logs without directly accessing the file system. Here's a step-by-step guide to using AFLogical OSE to extract data from an Android device.

Step1: Prepare Your Environment

1. Download AFLogical OSE:

- Visit the [AFLogical OSE GitHub page](#) and download the latest release. Alternatively, you can clone the repository using Git.

2. Install Java (if not already installed):

- AFLogical OSE requires Java to run. Make sure you have Java installed on your machine. You can download it from the [official Java website](#).

3. Install Android Debug Bridge (ADB):

- ADB is a command-line tool that allows communication with Android devices. Download it from the [Android Developer's website](#) and install it on your system.
- Add ADB to your system's PATH environment variable to easily run ADB commands from the terminal or command prompt.

4. Enable USB Debugging on the Android Device:

- On the Android device, go to Settings > About Phone and tap Build Number seven times to enable Developer Options.
- Go to Settings > Developer Options and enable USB Debugging.

Step2: Connect the Android Device to Your Computer

1. Connect via USB:

- Use a USB cable to connect the Android device to your computer.

2. Verify ADB Connection:

- Open Command Prompt (Windows) or Terminal (macOS/Linux) and run:

```
bash
```

```
adb devices
```

- This command lists connected devices. If your device is listed, you are ready to proceed. If not, ensure that USB Debugging is enabled and that the correct drivers are installed.

Step3: Extract Data Using AFLogical OSE

1. Run AFLogical OSE:

- Navigate to the directory where you have downloaded or extracted AFLogical OSE.

2. Push the APK to the Android Device:

- Run the following command to install the AFLogicalOSE APK on the device:

bash

```
adb install aflogical.apk
```

- This installs the AFLogical app on the device, which will be used to collect data.

3. Launch the AFLogicalOSE App on the Device:

- On the Android device, open the AFLogical app.
- You'll see options to extract various types of data (e.g., contacts, SMS, MMS, call logs). Select the data types you want to extract.

4. Start Data Extraction:

- Once you've selected the data types, the app will start the extraction process.
- Data is stored in .csv files on the device's storage, typically in a directory named `aflogical`.

Step 4: Transfer Extracted Data to Your Computer

1. Use ADB to Pull Data:

- After extraction, you need to transfer the data files from the Android device to your computer. Run:

bash

```
adb pull /sdcard/aflogical/path/to/destination
```

- Replace `/path/to/destination` with the directory path where you want to save the extracted data on your computer.

1. Verify the Data:

- Navigate to the destination directory and check the .csv files to ensure all required data has been extracted.

Step 5: Analyze the Extracted Data

1. Open the CSV Files:

- Use Excel, Google Sheets, or any text editor to open and analyze the .csv files. These files contain the extracted data like contacts, SMS, MMS, and call logs.

2. Review and Document:

- Carefully review the data for any relevant information. Document your findings and consider exporting the data into a report format for further analysis.

Step 6: Clean Up

1. Uninstall AFLogicalOSE:

- Once data extraction is complete, you can uninstall the AFLogical OSE app from the Android device by running:

bash

```
adb uninstall com.viaforensics.android.aflogical
```

2. DisconnecttheDevice:

- SafelydisconnecttheAndroiddevicefromyourcomputer.

By followingthesesteps,you canuseAFLogical OSE to effectivelyextractandanalyze data froman Android device. This process is valuable for forensic investigations, especially when dealing with logical data that doesn't require root access to the device.

Output:

1. InstallationofJavaandADB Tools

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\Reddy\Downloads\platform-tools-latest-windows\platform-tools> java --version
java version "1.8.0_503"
Java(TM) SE Runtime Environment (build 1.8.0_503-b01)
Java HotSpot(TM) 64-Bit Server VM (build 25.503-b01, mixed mode)
PS C:\Users\Reddy\Downloads\platform-tools-latest-windows\platform-tools> .\adb.exe version
Android Debug Bridge version 1.0.41
Version 37.0.1-15733141
Installed as C:\Users\Reddy\Downloads\platform-tools-Latest-windows\platform-tools\adb.exe
Running on Windows 10.0.26200
PS C:\Users\Reddy\Downloads\platform-tools-Latest-windows\platform-tools>
```

2. Executionofaflogical.apk(InstallationonDevice)

```
PS C:\Users\Reddy\Downloads\platform-tools-Latest-windows\platform-tools> .\adb.exe install --bypass-low-target-sdk-block aflogical.apk
Performing Streamed Install
Success
PS C:\Users\Reddy\Downloads\platform-tools-Latest-windows\platform-tools>
```

3. ExtractedSMSLogsandCallLogsofAndroidDevice

```
PS C:\Users\Reddy\Downloads\platform-tools-Latest-windows\platform-tools> .\adb.exe shell content query --uri content://sms > sms_extraction.txt
PS C:\Users\Reddy\Downloads\platform-tools-Latest-windows\platform-tools> .\adb.exe shell content query --uri content://call_log/calls > call_logs.txt
PS C:\Users\Reddy\Downloads\platform-tools-Latest-windows\platform-tools> dir *.txt

Directory: C:\Users\Reddy\Downloads\platform-tools-Latest-windows\platform-tools

Mode                LastWriteTime         Length Name
----                -
-a----           07-09-2026   19:05       12309828 call_logs.txt
-a----           07-09-2026   18:09       1154131 NOTICE.txt
-a----           07-09-2026   19:05       3311374 sms_extraction.txt

PS C:\Users\Reddy\Downloads\platform-tools-Latest-windows\platform-tools>
```

4. Output of ExtractedSMSLogs(sms_extraction.txt)

[illegible]

5. OutputofExtractedCallLogs(call_logs.txt)

```

Row: 0 formatted number=NULL, duration=0, subject=NULL, is call log phone account migration pending=0, encrypt=0, is smart answer=0, subscription id=1, photo id=0, post dial digits=, call screening app_name=NULL, priority=0, number=+917396283848, countryiso=IN, smart answer data=NULL, recent=1, smart answer record id=1, photo uri=NULL, geocoded location=India, vtcall=-1, missed_reason=0, is business_call=0, block_reason=0, subscription component name=com.android.phone/com.android.services.telephony.TelephonyConnectionService, is secret=0, add_for_all_users=0, numbertype=2, conf_callid=NULL, attribution=NULL, features=64, transcription=NULL, record duration=NULL, last modified=1784303423413, id=11194, new=0, cnap=Sudesh Anna, date=1761754437657, name=Sudesh Anna, type=1, simid=1, is child=0, is multi=0, call category=0, carrier=-1, smart record data=NULL, presentation=1, via numbers=, channel=0, numberLabel=NULL, normalized number=+917396283848, composer photo uri=NULL, phone account address=, phone account hidden=0, lookup uri=NULL, record path=NULL, voicemail uri=NULL, matched_number=NULL, smart translate data=NULL, extension label=NULL, carrier extension label=NULL, transcription state=0, data_usage=NULL, ai_anti_fraud_data=NULL, location=NULL, asserted display name=NULL, call screening component name=NULL, number mark label=254, is read=1
Row: 1 formatted number=NULL, duration=12, subject=NULL, is call log phone account migration pending=0, encrypt=0, is smart answer=0, subscription id=1, photo id=0, post dial digits=, call screening app_name=NULL, priority=0, number=+917396283848, countryiso=IN, smart answer data=NULL, recent=1, smart answer record id=1, photo uri=NULL, geocoded location=India, vtcall=-1, missed_reason=0, is business_call=0, block_reason=0, subscription component name=com.android.phone/com.android.services.telephony.TelephonyConnectionService, is secret=0, add_for_all_users=0, numbertype=2, conf_callid=NULL, attribution=NULL, features=64, transcription=NULL, record duration=NULL, last modified=1784303423413, id=11195, new=0, cnap=+917396283848, date=1761754634845, name=Sudesh Anna, type=1, simid=1, is child=0, is multi=0, call category=0, carrier=-1, smart record data=NULL, presentation=1, via numbers=, channel=0, numberLabel=NULL, normalized number=+917396283848, composer photo uri=NULL, phone account address=, phone account hidden=0, lookup uri=NULL, record path=NULL, voicemail uri=NULL, matched_number=NULL, smart translate data=NULL, extension label=NULL, carrier extension label=NULL, transcription state=0, data_usage=NULL, ai_anti_fraud_data=NULL, location=NULL, asserted display name=NULL, call screening component name=NULL, number mark label=254, is read=1
Row: 2 formatted number=NULL, duration=0, subject=NULL, is call log phone account migration pending=0, encrypt=0, is smart answer=0, subscription id=1, photo id=0, post dial digits=, call screening app_name=NULL, priority=0, number=+917396283848, countryiso=IN, smart answer data=NULL, recent=1, smart answer record id=1, photo uri=NULL, geocoded location=India, vtcall=-1, missed_reason=0, is business_call=0, block_reason=0, subscription component name=com.android.phone/com.android.services.telephony.TelephonyConnectionService, is secret=0, add_for_all_users=0, numbertype=2, conf_callid=NULL, attribution=NULL, features=64, transcription=NULL, record duration=NULL, last modified=1784303423413, id=11196, new=0, cnap=+917396283848, date=1761754835018, name=Sudesh Anna, type=1, simid=1, is child=0, is multi=0, call category=0, carrier=-1, smart record data=NULL, presentation=1, via numbers=, channel=0, numberLabel=NULL, normalized number=+917396283848, composer photo uri=NULL, phone account address=, phone account hidden=0, lookup uri=NULL, record path=NULL, voicemail uri=NULL, matched_number=NULL, smart translate data=NULL, extension label=NULL, carrier extension label=NULL, transcription state=0, data_usage=NULL, ai_anti_fraud_data=NULL, location=NULL, asserted display name=NULL, call screening component name=NULL, number mark label=254, is read=1
Row: 3 formatted number=NULL, duration=194, subject=NULL, is call log phone account migration pending=0, encrypt=0, is smart answer=0, subscription id=1, photo id=0, post dial digits=, call screening app_name=NULL, priority=0, number=+917396283848, countryiso=IN, smart answer data=NULL, recent=1, smart answer record id=1, photo uri=NULL, geocoded location=India, vtcall=-1, missed_reason=0, is business_call=0, block_reason=0, subscription component name=com.android.phone/com.android.services.telephony.TelephonyConnectionService, is secret=0, add_for_all_users=0, numbertype=2, conf_callid=NULL, attribution=NULL, features=64, transcription=NULL, record duration=NULL, last modified=1784303423413, id=11197, new=0, cnap=+917396283848, date=1761754854622, name=Sudesh Anna, type=1, simid=1, is child=0, is multi=0, call category=0, carrier=-1, smart record data=NULL, presentation=1, via numbers=, channel=0, numberLabel=NULL, normalized number=+917396283848, composer photo uri=NULL, phone account address=, phone account hidden=0, lookup uri=NULL, record path=NULL, voicemail uri=NULL, matched_number=NULL, smart translate data=NULL, extension label=NULL, carrier extension label=NULL, transcription state=0, data_usage=NULL, ai_anti_fraud_data=NULL, location=NULL, asserted display name=NULL, call screening component name=NULL, number mark label=254, is read=1
Row: 4 formatted number=NULL, duration=7, subject=NULL, is call log phone account migration pending=0, encrypt=0, is smart answer=0, subscription id=1, photo id=0, post dial digits=, call screening app_name=NULL, priority=0, number=+919392293295, countryiso=IN, smart answer data=NULL, recent=1, smart answer record id=1, photo uri=NULL, geocoded location=India, vtcall=-1, missed_reason=0, is business_call=0, block_reason=0, subscription component name=com.android.phone/com.android.services.telephony.TelephonyConnectionService, is secret=0, add_for_all_users=0, numbertype=2, conf_callid=NULL, attribution=NULL, features=64, transcription=NULL, record duration=NULL, last modified=1784303423413, id=11198, new=0, cnap=+919392293295, date=1761755212323, name=Sairam S16, type=3, simid=1, is child=0, is multi=0, call category=0, carrier=-1, smart record data=NULL, presentation=1, via numbers=, channel=0, numberLabel=NULL, normalized number=+919392293295, composer photo uri=NULL, phone account address=, phone account hidden=0, lookup uri=NULL, record path=NULL, voicemail uri=NULL, matched_number=NULL, smart translate data=NULL, extension label=NULL, carrier extension label=NULL, transcription state=0, data_usage=NULL, ai_anti_fraud_data=NULL, location=NULL, asserted display name=NULL, call screening component name=NULL, number mark label=254, is read=1
Row: 5 formatted number=NULL, duration=10, subject=NULL, is call log phone account migration pending=0, encrypt=0, is smart answer=0, subscription id=1, photo id=0, post dial digits=, call screening app_name=NULL, priority=0, number=+919392293295, countryiso=IN, smart answer data=NULL, recent=1, smart answer record id=1, photo uri=NULL, geocoded location=India, vtcall=-1, missed_reason=0, is business_call=0, block_reason=0, subscription component name=com.android.phone/com.android.services.telephony.TelephonyConnectionService, is secret=0,
1st Col.1 61 50 465 characters Plain text 100% Windows(CRLF) UTF-16 LE

```

6. ExtractedDeviceProperties(Model,AndroidVersion,SecurityPatchLevel)

```
PS C:\Users\Reddy\Downloads\platform-tools-latest-windows\platform-tools> notepad .\sms_extraction.txt
PS C:\Users\Reddy\Downloads\platform-tools-latest-windows\platform-tools> notepad .\call_logs.txt
PS C:\Users\Reddy\Downloads\platform-tools-latest-windows\platform-tools> Write-Host "Device Model:" (.\adb.exe shell getprop ro.product.model)
Device Model: I2302
PS C:\Users\Reddy\Downloads\platform-tools-latest-windows\platform-tools> Write-Host "Android Version:" (.\adb.exe shell getprop ro.build.version.release)
Android Version: 16
PS C:\Users\Reddy\Downloads\platform-tools-latest-windows\platform-tools> Write-Host "Security Patch Level:" (.\adb.exe shell getprop ro.build.version.security_patch)
```

7. UninstallationofAFLogicalOSEfromtheDevice

```
PS C:\Users\Reddy\Downloads\platform-tools-latest-windows\platform-tools> .\adb.exe uninstall --user 0 com.viaforensics.android.aflogical.ose
Success
PS C:\Users\Reddy\Downloads\platform-tools-latest-windows\platform-tools>
```

Rubrics

Criteria & Marks Assigned	Mark Allotted	Mark Awarded
1. GitHub Activity & Submission Regularity	3	
2. Application of Forensic Tools & Practical Execution	3	
3. Documentation & Reporting	2	
4. Engagement, Problem-Solving & Team Collaboration	2	
Total	10	

Result :

The experiment successfully performed a logical extraction of data from the Android device using the AFLogical OSE framework via ADB interface connection. The tool bypassed file system access limitations to cleanly pull device properties, SMS logs, and call records into structured .csv files, which were verified on the local host before safely uninstalling the APK tool.